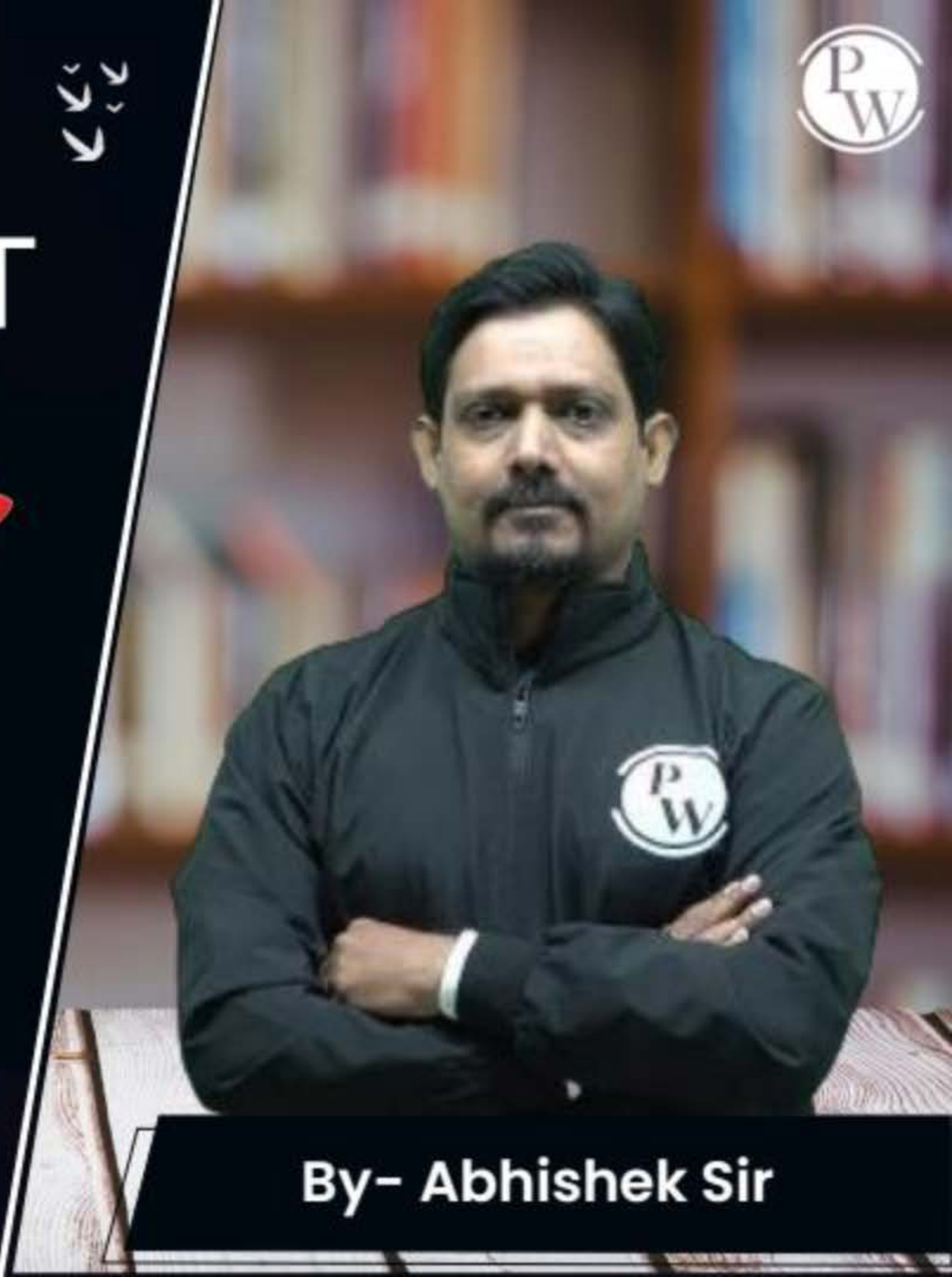
Computer Science & IT



Fundamental of C Language

Control Flow Statement

Lecture No. 03



By- Abhishek Sir

Recap of Previous Lecture



- Topic
- Topic
- Topic
- Topic
- Topic

Switch statement

for Loop

Slide

Topics to be Covered



- Topic — while loop
- Topic — do while loop
- Topic — break
- Topic — continue
- Topic

Slide



Loops

Iterative statements : while loop

do while loop



While loop

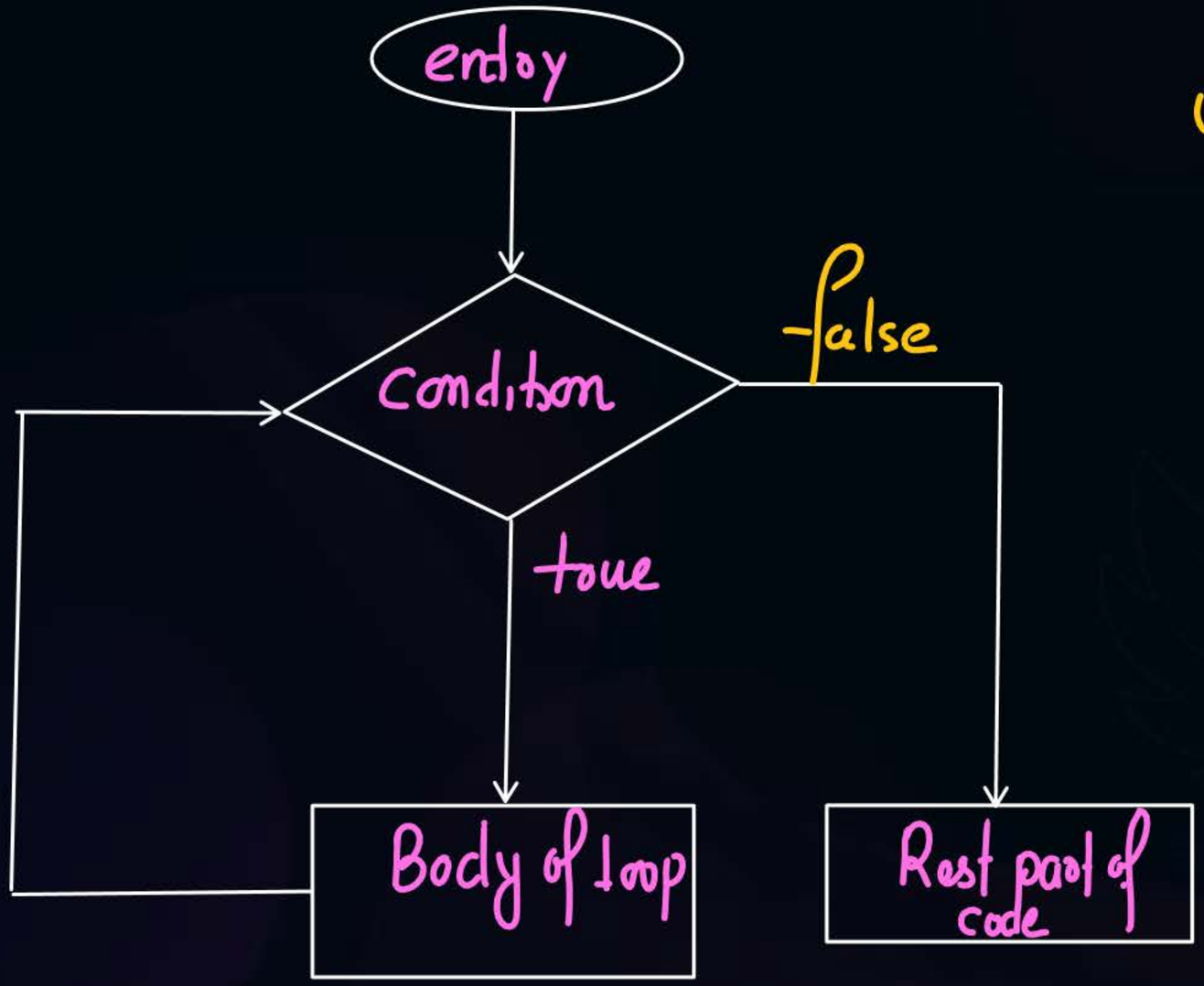
While loop entry controlled loop

```
while (condition) {

}
```

* No of iteration is predefined in for loop
* while loop is condition dependent

While loop



while (condition)

↑

Mandatory



While loop

```
# include<stdio.h>
int main() {
    int a=10,
    while(a<=20) {
        printf("Name");
        a=a+2;
    }
    return 0;
}
```

Name will be printed 6 times

10<=20,	Name	12
12<=20,	Name	14
14<=20	Name	16
16<=20	Name	18
18<=20	Name	20
20<=20	Name	22

22<=20

Slide



Do-while Loop

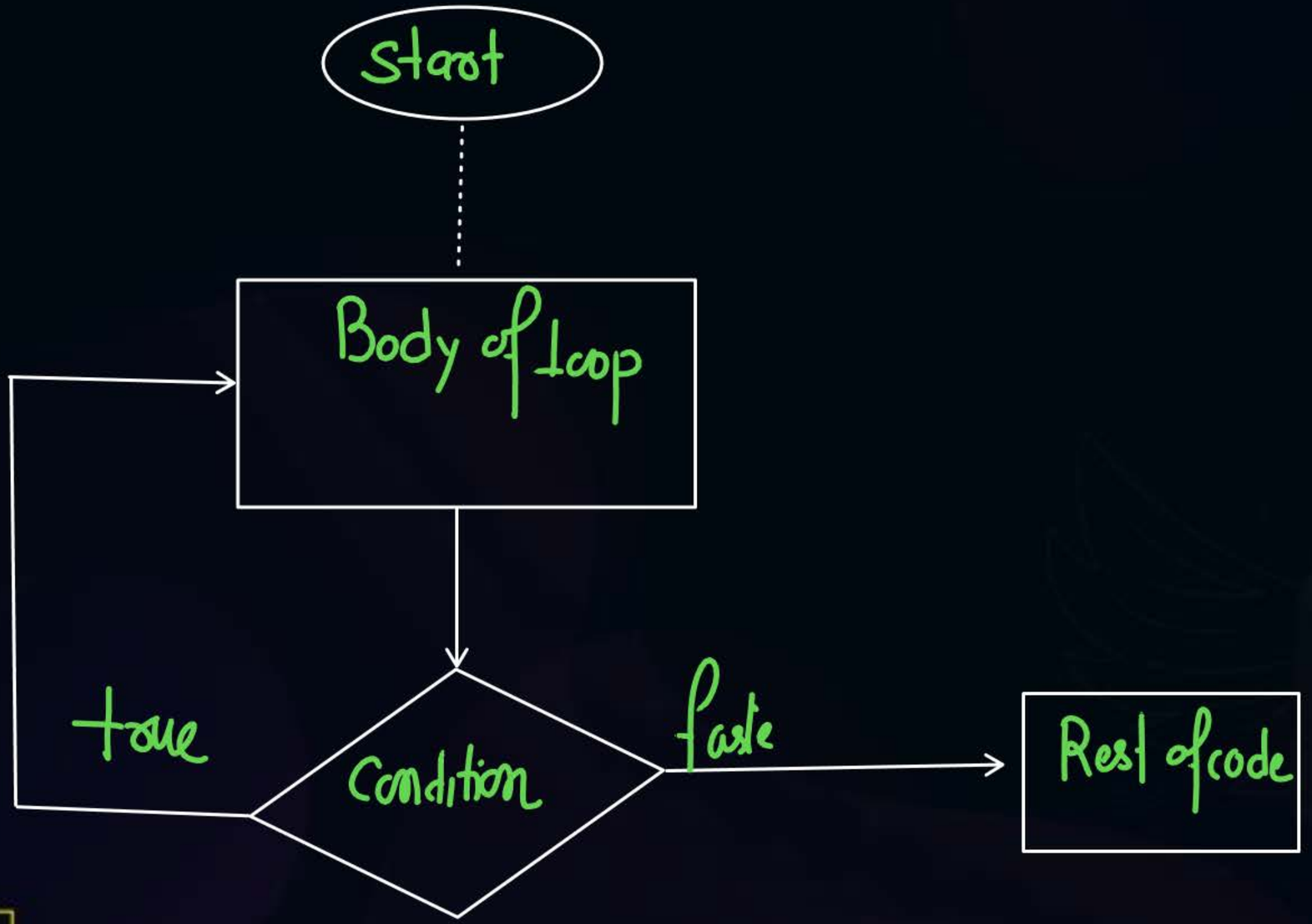
1. do while loop, first block will execute
   then condition will be tested
2. exit controlle loop
3. At least one time body of loop will execute
   even if condition is false

```
do {
} while(condition);
```

Slide



Do-while Loop





Do-while Loop

```
# include<stdio.h>
int main() {
    int a = 4;
    do {
        printf("Name");
        a = a-1;
    } while (a>=0);
    return 0;
}
```

Name	3	3>=0
Name	2	2>=0
Name	1	1>=0
Name	0	0>=0
Name	-1	-1>=0 false

Slide



break

Loop control statement :- break statement cause control to transfer out of switch or loop block in which break is used.

Slide



break

```
#include<stdio.h>
int main() {
   int i;

   for(i=1; i<=10,i++) {
         printf("Name");
         if(a==4) break;
   }
   printf("outside break");
   return 0;
}
```

1<=10 Name 1==4

2<=10 Name 2==4

3<=10 Name 3==4

4<=10 Name 4==4 break.

outside break

Slide



break

```
# include<stdio.h>
int main() {
   int i;

   for (i=1; i<=10, i++) {
         printf ("Name");
         break;
   }
   printf ("outside break");
   return 0;
}
```

1<=10 Name break

outside break

Slide



break

break will work with do while while also

Switch and Loop



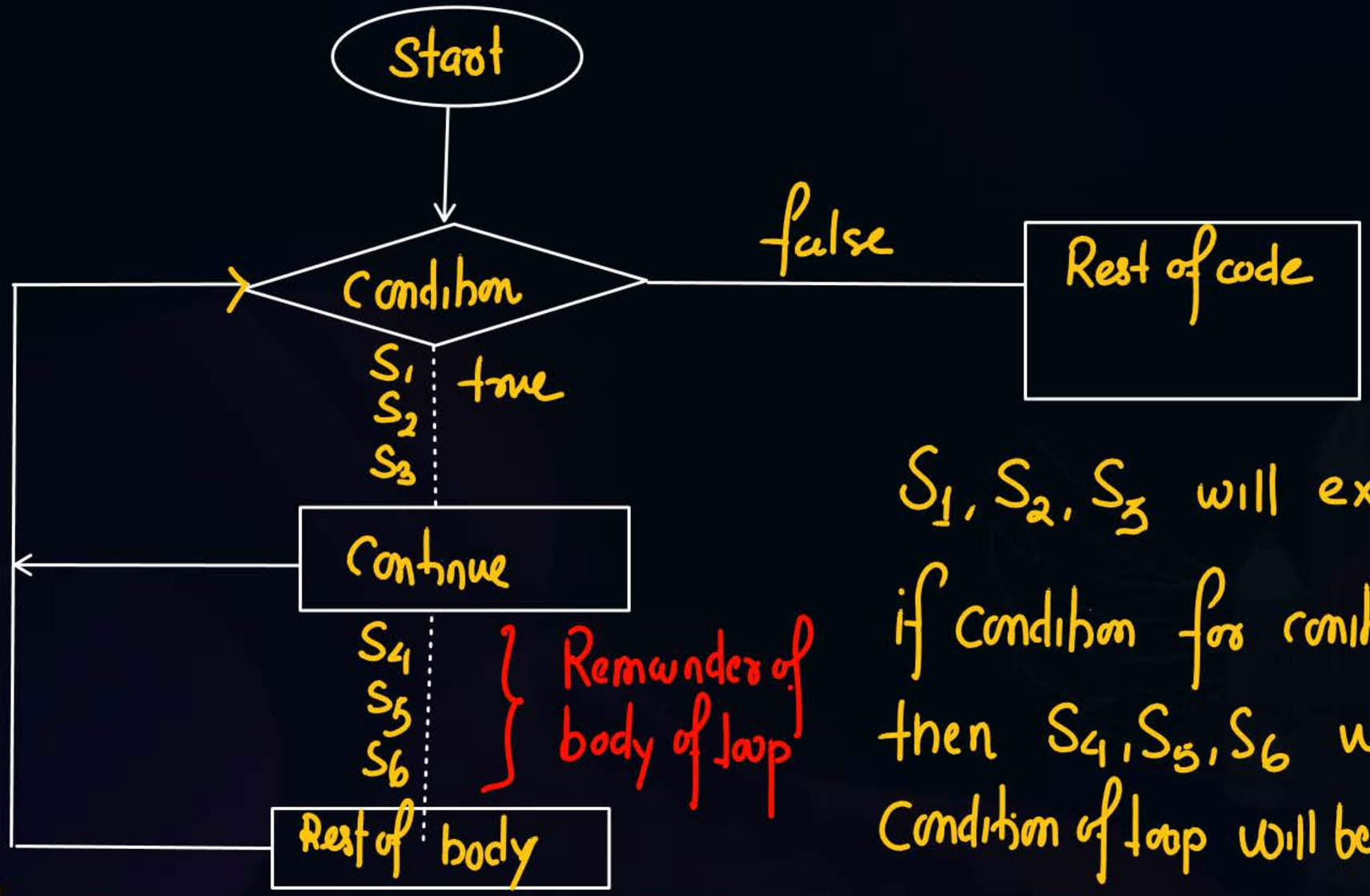
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Continue

* Continue does not break the loop
* Continue cause skip remainder of body of loop and retest the condition before reiterating

Continue



S_1, S_2, S_3 will execute if condition for continue is true then S_4, S_5, S_6 will be skip and Condition of loop will be tested

Slide



Continue

```
# include<stdio.h>
int main() {
    int i=1,
    while ( i<=10) {
        if (i>4) continue;
        printf (" Name");
        i=i+2;
    }
    return 0;
}
```

Infinite Loop

Condition	Condition for continue	print	Increm
1<=10	1>4	Name	3
3<=10	3>4	Name	5
5<=10	5>4	—	—
5<=10	5>4	—	—
5<=10	5>4	—	—
⋮	⋮		

Slide



2 mins Summary

Topic — while Loop

Topic — do - while

Topic — break

Topic — contine

Topic

Slide

THANK - YOU

